Lab Documentation: freeCodeCamp — Build a Cat Painting

Project Name: Build a Cat Painting

Certification: Responsive Web Design Certification

Platform: freeCodeCamp

1. Overview

This project involved constructing pure CSS artwork—a cat portrait—from scratch using semantic HTML5 structuring and advanced CSS3 styling techniques. The primary objective was to leverage CSS box model properties, absolute positioning, transformations, and custom borders to form geometric facial components without external image assets.

2. Technical Workflow & Implementation

A. Document Setup & CSS Reset

- Established standard HTML boilerplate with `<meta charset="UTF-8">`, stylesheet linking, and standard semantic markup.
- Applied a universal CSS reset using `box-sizing: border-box` to ensure consistent dimensions across elements.
- Applied background color styling (`#c9d2fc`) across the `body` container.

B. Structural Layout & Absolute Positioning Context

- Created nested HTML `div` blocks representing facial features (`.cat-head`, `.cat-ears`, `.cat-eyes`, `.cat-nose`, `.cat-mouth`, `.cat-whiskers`).
- Positioned `.cat-head` using centered absolute positioning (`position: absolute; top: 0; right: 0; bottom: 0; left: 0; margin: auto;`).
- Styled `.cat-head` with a subtle linear gradient (`linear-gradient(#5e5e5e 85%, #45454f 100%)`) and border curvature.

C. Constructing Features Using CSS Geometry

1. **Ears (`.cat-left-ear`, `.cat-right-ear`):**
    - Utilized transparent lateral borders (`border-left: 35px solid transparent`, `border-right: 35px solid transparent`) combined with a solid bottom border (`70px solid #5e5e5e`) to form triangular ear geometries.

    - Applied `transform: rotate()` to angle the ears (-45 deg and 45 deg) and manage stacking layers via `z-index: 1`.

    - Added inner ears using smaller border dimensions and rounded corners (`border-radius`).

2. **Eyes (`.cat-left-eye`, `.cat-right-eye`, Inner Pupils):**

    - Created elliptical shapes via precise height/width ratios and asymmetric `border-radius: 60%`.

    - Rotated the eyes outward (25 deg and -25 deg) for realistic cat eye geometry.

    - Placed inner pupils (`.cat-left-inner-eye`, `.cat-right-inner-eye`) using precise `top`/`left` offsets, white backgrounds (`#fff`), and slight rotations.

3. **Nose & Mouth (`.cat-nose`, `.cat-mouth-line-left`, `.cat-mouth-line-right`):**

    - Formed a triangular nose using asymmetrical border radii (`border-top-left-radius: 50%`, etc.).

    - Built curved mouth lines using absolute positioning, custom rotation angles (`rotate(170deg)` and `rotate(165deg)`), and precise offset placements.

3. Knowledge Gained & Technical Takeaways

- **CSS Border Triangle Trick:** Learned how combining zero width/height with transparent left/right borders and a solid top/bottom border creates clean vector triangles in pure CSS.

- **Asymmetrical Border Radius:** Mastered individual corner rounding (`border-top-left-radius`, `border-bottom-right-radius`) to craft complex organic shapes such as ears, eyes, and noses.

- **Layering & Precise Placement:** Reenforced the interaction between parent-child `position: absolute` contexts, coordinate offsets (`top`, `left`, `right`, `bottom`), `z-index` layering, and 2D rotations via `transform: rotate()`.

CODE:


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Build a Cat Painting
Instructions
index.html
styles.css
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index.html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>fCC Cat Painting</title>
<link rel="stylesheet" href="./styles.css">
</head>
<body>
<main>

styles.css
* {
box-sizing: border-box;
}
body {
background-color: #c9d2fc;
}
.cat-head {
position: absolute;
right: 0;
left: 0;
top: 0;
bottom: 0;
margin: auto;
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styles.css
.cat-head {
margin: auto;
background: linear-gradient(#5e5e5e 85%,
#45454f 100%);
width: 205px;
height: 180px;
border: 1px solid #000;
border-radius: 46%;
}
.cat-left-ear {
position: absolute;
top: -26px;
left: -31px;
z-index: 1;
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div>
</
div>
styles.css
.cat-left-ear {
z-index: 1;
border-top-left-radius: 90px;
border-top-right-radius: 10px;
transform: rotate(-45deg);
border-left: 35px solid transparent;
border-right: 35px solid transparent;
border-bottom: 70px solid #5e5e5e;
}
.cat-right-ear {
position: absolute;
top: -26px;
left: 163px;
z-index: 1;
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div>
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div>
styles.css
.cat-right-ear {
transform: rotate(45deg);
border-top-left-radius: 90px;
border-top-right-radius: 10px;
border-left: 35px solid transparent;
border-right: 35px solid transparent;
border-bottom: 70px solid #5e5e5e;
}
.cat-left-inner-ear {
position: absolute;
top: 22px;
left: -20px;
border-top-left-radius: 90px;
border-top-right-radius: 10px;
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div>
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div>
styles.css
.cat-left-inner-ear {
border-bottom-right-radius: 40%;
border-bottom-left-radius: 40%;
border-left: 20px solid transparent;
border-right: 20px solid transparent;
border-bottom: 40px solid #3b3b4f;
}
.cat-right-inner-ear {
position: absolute;
top: 22px;
left: -20px;
border-top-left-radius: 90px;
border-top-right-radius: 10px;
border-bottom-right-radius: 40%;
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div>
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div>
styles.css
.cat-right-inner-ear {
border-bottom-left-radius: 40%;
border-left: 20px solid transparent;
border-right: 20px solid transparent;
border-bottom: 40px solid #3b3b4f;
}
.cat-left-eye {
position: absolute;
top: 54px;
left: 39px;
border-radius: 60%;
transform: rotate(25deg);
width: 30px;
height: 40px;
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div>
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div>
styles.css
.cat-left-eye {
background-color: #000;
}
.cat-right-eye {
position: absolute;
top: 54px;
left: 134px;
border-radius: 60%;
transform: rotate(-25deg);
width: 30px;
height: 40px;
background-color: #000;
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div>
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div>
styles.css
.cat-left-inner-eye {
position: absolute;
top: 8px;
left: 2px;
width: 10px;
height: 20px;
transform: rotate(10deg);
background-color: #fff;
border-radius: 60%;
}
.cat-right-inner-eye {
position: absolute;
top: 8px;
left: 18px;
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div>
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div>
styles.css
.cat-right-inner-eye {
transform: rotate(-5deg);
width: 10px;
height: 20px;
background-color: #fff;
border-radius: 60%;
}
.cat-nose {
position: absolute;
top: 108px;
left: 85px;
border-top-left-radius: 50%;
border-bottom-right-radius: 50%;
border-bottom-left-radius: 50%;
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div>
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div>
styles.css
.cat-mouth-line-left {
position: absolute;
top: 88px;
left: 74px;
transform: rotate(170deg);
}
.cat-mouth-line-right {
position: absolute;
top: 88px;
left: 91px;
transform: rotate(165deg);
}
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